Anirudh Garg

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Education

New York University - Courant Institute of Mathematical Sciences Aug 2023 – May 2025
M.S. in Computer Science
Indian Institute of Technology, Kanpur July 2017 – May 2021
B.Tech. in Electrical Engineering

Technical Skills

- **Courses:** Big Data Processing, Deep Learning, NLP, Machine Learning, Distributed Systems, Algorithms
- **Programming Skills:** Python, PyTorch, SQL, Spark, Hadoop, MapReduce, Trino, C++, CUDA, GPU, Git

Work Experience

- **Software Engineer - Google, Mountain View, CA, USA** (*Oct '25 – Present*)
  - Optimized the **Learned Ranking Function** for **YouTube Shorts** by engineering new behavioral features, resulting in increased user retention and improved content discovery across the highly diverse global platform
  - Conducted rapid A/B testing on ranking models, refining the balance between exploration and exploitation.
- **Software Engineer - Amazon, Dallas, TX, USA** (*June '25 – Oct '25*)
  - Implemented migration of internal pipelines from **JDK 8 to JDK 17**, which improved performance, resolved backward compatibility issues, managed dependencies,e and addressed runtime changes across services
  - Refactored codebase to adopt **JDK 17 features**, and coordinated with teams for a smooth production rollout
- **Machine Learning Engineer - Samsung Research, Bangalore, India** (*July '21 – July '23*)
  - Developed a **few-shot** learning classification framework to assess Bixby Voice Assistant's performance across **35 different capsules**, classifying user sessions using **data augmentation** techniques and **attention** models
  - Engineered unified attention model, reducing processing time by **97%** and achieving **90%** classification accuracy
- **Summer Intern - Samsung Research, Bangalore, India** (*May '20 – July '20*)
  - Trained and optimized multiple recommendation models for the Bixby Voice Assistant with an accuracy of **87%**
  - Benchmarked performances of the classification models for the recommendation system for Bixby Voice Assistant

Research Publication

- A Semi-Supervised Approach for Multi-Domain Classification - **Anirudh Garg**, Kartikey Singh, Radhika Mundra
  (Accepted and published) [ICEET 2023]

Projects

- **EVALUATING LARGE LANGUAGE MODELS FOR EFFICIENT QA SYSTEMS** (*Sept '24 – Dec '24*)
  Mentor: Prof. He He - Computer Science Department, NYU CDS [Report] [Code]
  - Conducted a detailed study of BERT, T5 and Mamba models on QA tasks, balancing EM scores and efficiency
  - Explored advanced tuning techniques like LoRA (Low Rank Adaptation), Quantization and QLoRA for T5, achieving a **75%** reduction in memory usage and **60%** drop in training time with minimal drop in EM score
- **DISTRIBUTED FINANCIAL RISK ASSESSMENT** (*Sept '24 – Dec '24*)
  Mentor: Prof. Yang Tang - Computer Science Department, NYU Courant [Report] [Code]
  - Developed a scalable financial risk assessment system using **Apache Spark MLlib** and **Monte Carlo** simulations
  - Simulated **100,000** portfolio return trials, quantified risks, and delivered robust decision-making financial metrics
- **EMERGENCY RESPONSE EFFICIENCY ANALYSIS** (*Sep '24 – Dec '24*)
  Mentor: Prof. Yang Tang - Computer Science Department, NYU Courant [Report] [Code]
  - Analyzed San Francisco fire and EMS response times, using **HDFS MapReduce** and **Trino** to improve safety
- **FUTURE VIDEO FRAME SEGMENTATION PREDICTION** (*Sept '23 – Dec '23*)
  Mentor: Prof. Yann LeCun - Computer Science Department, NYU Courant [Report] [Code]
  - Employed SimVP and Vision Transformer models with diverse object attributes to predict the segmentation
  - Used the first 11 frames of a video to achieve accurate prediction for the 22nd frame with an accuracy of **40%**
- **VARIABLE SIZED-BATCH GENERAL MULTIPLICATION (VARIABLE GEMM)** (*Sept '23 – Dec '23*)
  Mentor: Prof. Jinyang Li - Computer Science Department, NYU Courant [PPT] [Code]
  - Improved load-balancing in Mixture-of-Experts using CuBLAS and CuSparse, enhancing GPU/TPU efficiency
  - Implemented MoE layer with CuBLAS for non-uniform GEMM, achieving faster, parallelized computations

Scholastic Achievements

- Conferred with a merit certificate at **2016 Indian National Math Olympiad**, ranked among top 50 students
- Awarded with the prestigious **KVPY** fellowship in 2016 with an **All India Rank 370** among 40,000 students